Investigation into UFO Sightings Across US

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Hypotheses

Upon initial inspection of the data, we proposed three hypotheses for investigation.

- How does geographical location correlate with the frequency of UFO sightings across the US?
- How has the number of UFO sightings in the US changed over time? Why?
- How do the four seasons differ in the proportion of UFO sightings they account for?

Data Wrangling

The data set contains 88125 UFO sightings from 1969 to 2019. Before we start our analysis, we conduct some data cleaning on the data.

- Removing irrelevant variables.
  - Url, text data, duration, etc...
- Converting timelogs to month, day, year.
- Matching up longitude, latitude, city, and state given there were some false entries.
- Creating a new variable named "season" using months

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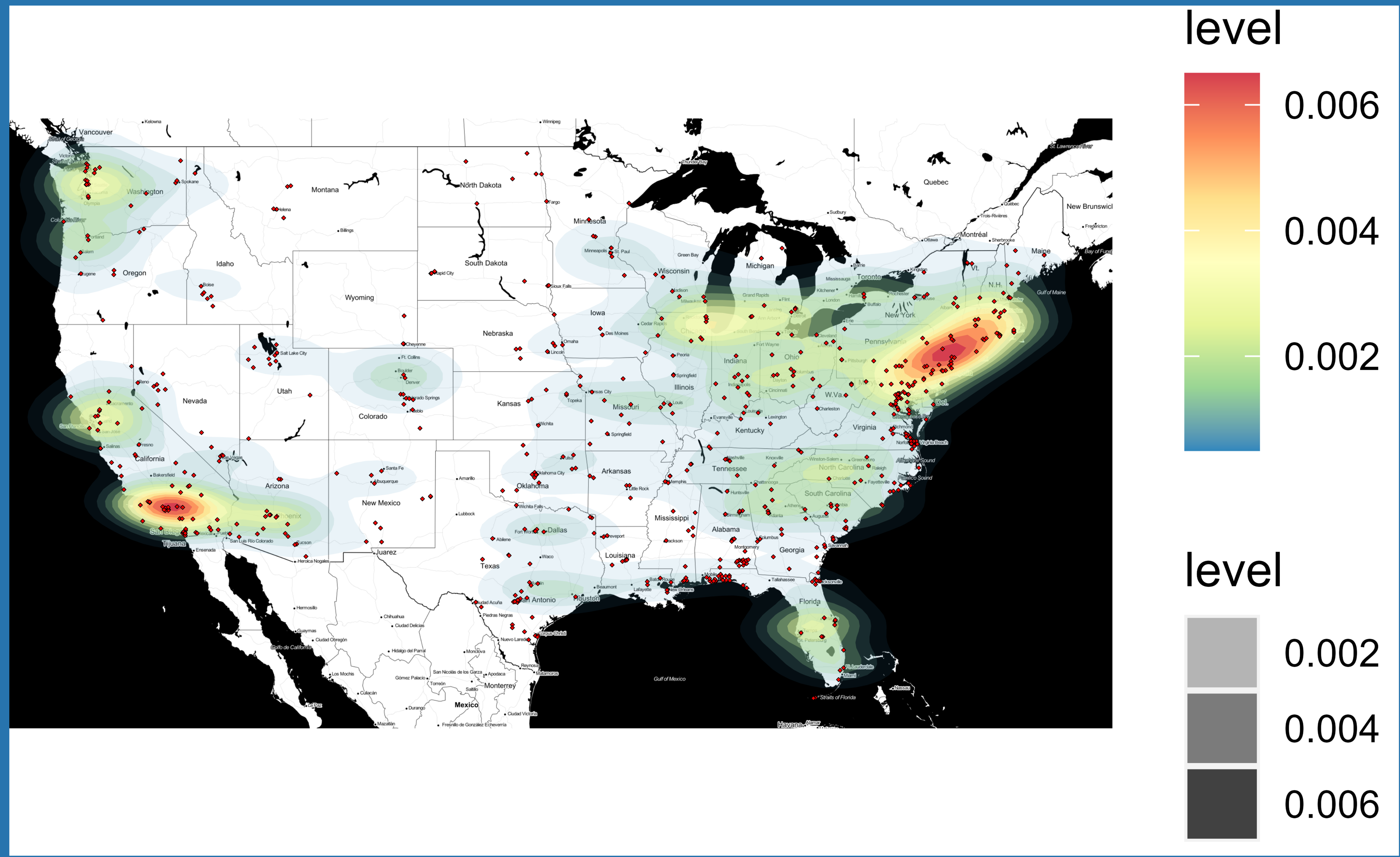
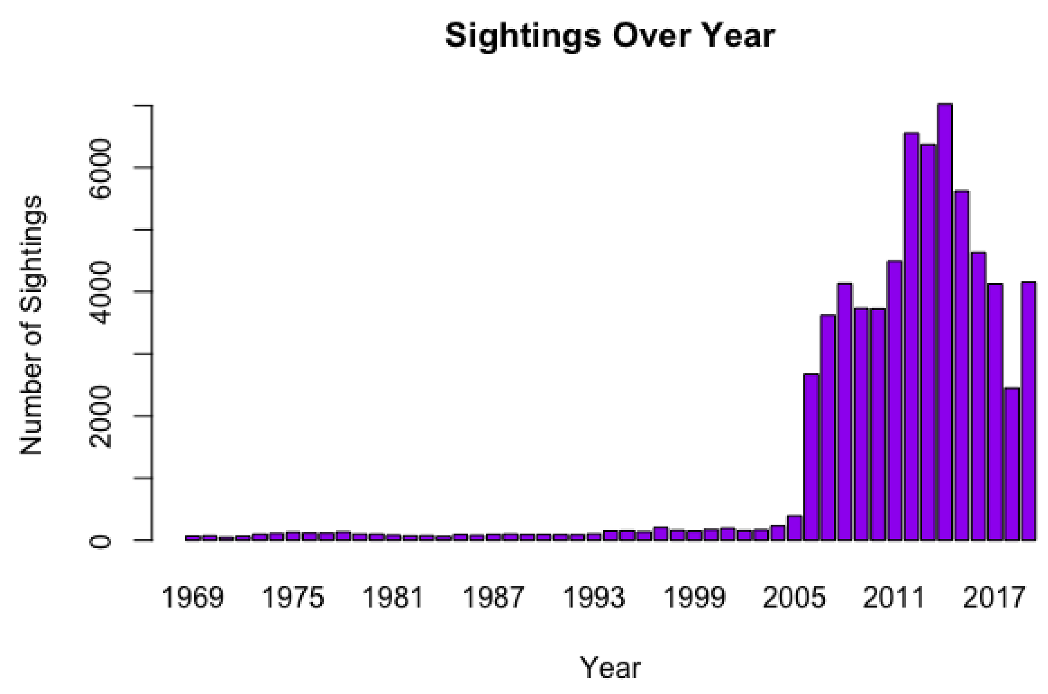
Heat Map of Sightings Across the US



- Only 4.3% of observations came from Hawaii, Alaska, and Canada
- The more military bases there were in an area the greater the sightings
- The larger the city the more sightings were observed

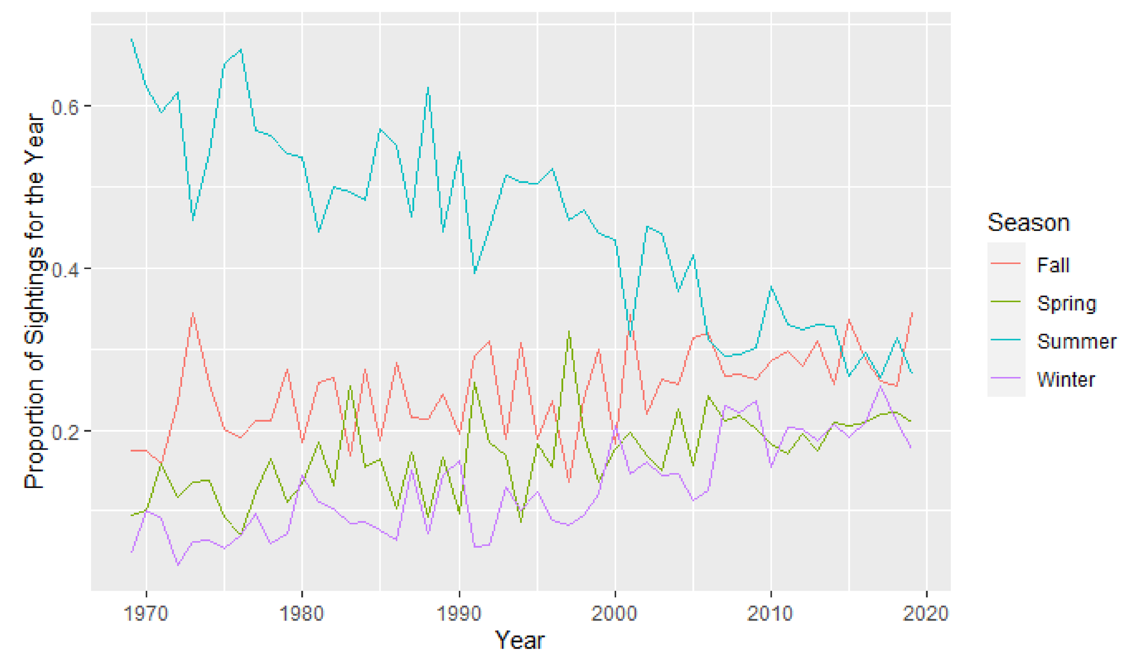
UFO Sightings Over Time



Findings:

- Sudden spike in 2006.
- Increase in variation post-2006.

UFO Sightings Over Season



Findings:

- Much more effective than the previous plot
- There **IS** an interaction between season and time
- Effects of seasons grew to be similar over the course of time
- **Summer was largely dominant at first but for some reason regressed to the mean**

Further Research

To better understand the observed patterns in UFO sightings, more analysis is required. Researchers could investigate changes in reporting practices or the public interest, and gather more detailed data on locations and times to uncover the underlying factors.